F1+TF1 COMPUTER NETWORKS

Case Study 1 | Literature Survey

CoWiC-MAC: A Coordinated WiFi-Centric MAC Mechanism in Heterogeneous Wireless Networks

A Literature Survey Based on the Base Paper

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2. Base Paper Identification

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Based on the base paper's own reference list, CoWiC-MAC belongs to the Cross-Technology Communication (CTC) and WiFi/low-power-radio coexistence research line. It addresses coordinated medium access between WiFi and heterogeneous low-power wireless devices, such as ZigBee/IEEE 802.15.4-class radios, sharing the 2.4 GHz ISM band, building on the authors' own prior CTC work (WibZig, IPSN 2023; WAIC-CTC, Computer Networks 2025). The literature survey below is therefore scoped to WiFi-centric MAC design, WiFi/ZigBee coexistence, cross-technology communication, and coordinated or explicit channel-access mechanisms, including IEEE 802.11ax high-efficiency MAC features, since these are the technical neighborhoods the base paper itself draws on.

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3. Literature Survey Table

S.No	Title of the Paper, Author Name, Year	Methodology	Key Findings	Advantage and Limitations
1	Performance Analysis of the IEEE 802.11 Distributed Coordination Function, G. Bianchi, 2000	Develops a two-dimensional discrete-time Markov chain model of the 802.11 DCF binary exponential backoff process to analytically compute saturation throughput for both the Basic Access and RTS/CTS access mechanisms under an ideal channel and a finite number of contending terminals.	The analytical model matches simulation results with high accuracy across both access mechanisms; RTS/CTS is shown to outperform Basic Access as the number of contending stations and payload size grow, since it reduces the cost of collisions.	**Advantages:** Provides the first rigorous, closed-form throughput model for CSMA/CA-based WiFi MAC, later reused and extended by almost all subsequent 802.11 MAC performance studies. **Limitations:** Assumes saturated traffic and a homogeneous single-BSS network; does not model heterogeneous coexisting technologies or non-saturated, bursty traffic (*inferred limitation*).
2	PPR: Partial Packet Recovery for Wireless Networks, K.	Proposes a PHY/link-layer cross-layer scheme in which the receiver identifies which chip/bit segments of a corrupted	Partial retransmission substantially reduces the airtime wasted on full-frame retransmissions under lossy	**Advantages:** Fine-grained recovery reduces MAC-layer retransmission overhead without requiring changes to the higher-

S.No	Title of the Paper, Author Name, Year	Methodology	Key Findings	Advantage and Limitations
	Jamieson & H. Balakrishnan, 2007	802.11 frame were received correctly, using PHY confidence information, and requests retransmission of only the corrupted segments instead of discarding and resending the whole frame.	or interfering channel conditions compared with standard ARQ, improving effective throughput as loss rates increase.	layer protocol stack. **Limitations:** Requires PHY-layer chip/bit-confidence estimation support in the receiver hardware, adding implementation complexity; benefits diminish under very low error rates (*inferred limitation*).
3	Beyond Co-existence: Exploiting WiFi White Space for ZigBee Performance Assurance, J. Huang, G. Xing, G. Zhou, R. Zhou, 2010	Introduces a scheme that senses and predicts idle temporal white spaces between WiFi transmissions and schedules ZigBee (IEEE 802.15.4) transmissions into those gaps, without requiring any modification to WiFi devices.	Scheduling ZigBee transmissions into predicted WiFi-idle intervals substantially improves ZigBee packet delivery under heavy co-channel WiFi traffic compared to unmanaged coexistence.	**Advantages:** Fully WiFi-transparent; no protocol or hardware change on the WiFi side is needed. **Limitations:** Prediction accuracy degrades under bursty, unpredictable WiFi traffic, and the scheme performs one-way, WiFi-unaware coordination rather than bidirectional negotiation (*inferred limitation*).
4	Surviving Wi-Fi Interference in Low Power ZigBee Networks, C.-J. M. Liang, N. B. Priyantha, J. Liu, A. Terzis, 2010	Presents a forward-error-correction coding scheme tailored to the burst-loss pattern caused by WiFi interference on ZigBee frames, adding structured redundancy so a ZigBee receiver can recover packets despite partial WiFi-induced corruption.	The coding scheme yields markedly higher ZigBee packet reception rates than uncoded transmission under representative WiFi interference conditions, without requiring cooperation from WiFi devices.	**Advantages:** A purely receiver/transmitter-side solution on the ZigBee side, deployable without touching WiFi infrastructure. **Limitations:** Added redundancy reduces effective ZigBee payload throughput and energy efficiency; the approach tolerates interference rather than eliminating it (*inferred limitation*).
5	Enabling Coexistence of Heterogeneous Wireless Systems: Case for ZigBee and WiFi, X. Zhang & K. G. Shin, 2011	Proposes a coexistence mechanism enabling WiFi and ZigBee devices to detect each other's channel activity and align their transmissions to avoid mutual interference in the shared 2.4 GHz band, using PHY-layer activity/energy detection rather than full protocol interoperability.	Cross-technology activity detection and alignment substantially reduces collision-induced packet loss for both ZigBee and WiFi relative to uncoordinated coexistence.	**Advantages:** Establishes an early foundation for cross-technology coexistence without requiring a shared MAC protocol between heterogeneous radios. **Limitations:** Coordination is implicit and detection-based rather than explicit signaling, limiting achievable precision and scalability beyond two technologies (*inferred limitation*).
6	WiCop: Engineering WiFi Temporal White-Spaces for Safe Operations of Wireless Personal Area Networks in Medical Applications, Y. Wang, Q. Wang, G. Zheng, Z. Zeng, R. Zheng, Q. Zhang, 2014	Passively monitors WiFi traffic to model its temporal occupancy pattern and schedules safety-critical WBAN (802.15.4-class) transmissions into predicted white spaces, with worst-case latency guarantees for medical sensor traffic.	Demonstrates bounded, predictable worst-case access latency for medical WBAN traffic coexisting with unmodified WiFi networks, reducing collision-induced delay and loss relative to unmanaged coexistence.	**Advantages:** Provides deterministic latency bounds suitable for safety-critical medical applications, a stronger guarantee than best-effort coexistence schemes. **Limitations:** Relies on accurate short-term WiFi traffic prediction, which can degrade in highly dynamic, multi-AP, or high-load environments (*inferred limitation*).
7	Enhancing ZigBee Throughput under WiFi Interference Using Real-Time Adaptive Coding, P. Guo, J. Cao, K. Zhang, X. Liu, 2014	Proposes an adaptive forward error correction scheme that dynamically adjusts code redundancy and rate in real time based on online estimation of WiFi interference intensity, aiming to maximize ZigBee throughput while preserving reliability.	Adaptive coding is shown to outperform both static-redundancy coding and uncoded transmission across varying WiFi traffic loads, improving ZigBee throughput while maintaining reliability targets.	**Advantages:** Interference-aware adaptation avoids the throughput penalty of always using worst-case, high redundancy. **Limitations:** Requires continuous interference estimation, adding computational and energy overhead on resource-constrained ZigBee nodes (*inferred limitation*).
8	Taming Cross-Technology Interference for Wi-Fi and ZigBee Coexistence Networks, P. Yang, Y. Yan, X.-Y. Li, Y. Zhang, Y. Tao, L. You, IEEE Trans. Mob. Comput., 2016	Builds an empirical, statistical model of cross-technology interference between WiFi and ZigBee and develops a mitigation strategy combining interference-aware channel/timing selection with adaptive transmission, addressing both detection and avoidance of cross-technology collisions.	The proposed interference model is validated experimentally, and the mitigation strategy achieves substantially improved ZigBee packet reception ratio and reduced retransmissions under realistic WiFi interference.	**Advantages:** Combines interference characterization with active mitigation rather than passive tolerance alone, improving robustness. **Limitations:** Model calibration is specific to the studied WiFi/ZigBee interference regime and may require retuning for other technology pairs or channel conditions (*inferred limitation*).
9	IEEE 802.11ax: High-Efficiency WLANs, B. Bellalta, IEEE	Surveys the MAC/PHY mechanisms introduced by the then-draft IEEE 802.11ax amendment, namely OFDMA-based multi-user access, MU-MIMO,	Identifies OFDMA scheduling and spatial reuse as the key mechanisms expected to raise aggregate WLAN efficiency in dense, heterogeneous	**Advantages:** Provides an accessible, comprehensive overview consolidating the new 802.11ax MAC toolbox relevant to coordinated channel access design.

S.No	Title of the Paper, Author Name, Year	Methodology	Key Findings	Advantage and Limitations
	Wireless Commun., 2016	spatial reuse via BSS coloring, and longer OFDM symbols, aimed at improving efficiency in dense, overlapping-BSS deployments.	deployments, while highlighting legacy-device coexistence as an open challenge.	**Limitations:** A tutorial/survey rather than a novel protocol; reports no new experimental results of its own (*inferred limitation*).
10	WEBee: Physical-Layer Cross-Technology Communication via Emulation, Z. Li & T. He, MobiCom, 2017	Introduces physical-layer emulation CTC, in which a high-speed WiFi OFDM radio manipulates only the payload of its packets to emulate the desired waveform of a low-speed radio such as ZigBee, enabling direct WiFi-to-ZigBee communication with no hardware or firmware changes on either side.	Achieves more than 99% reliable parallel CTC between WiFi and ZigBee at 126 Kbps in noisy environments, about 16,000 times faster than prior packet-level-modulation CTC schemes.	**Advantages:** Zero-cost, commodity-compatible deployment with orders-of-magnitude higher CTC throughput than earlier schemes; forms the technical basis for later PHY-CTC coordination work, including the base paper's own prior WibZig protocol. **Limitations:** Emulation fidelity depends on precise timing and waveform control at the WiFi transmitter and was validated primarily on the WiFi-to-ZigBee direction.
11	IEEE 802.11ax: Highly Efficient WLANs for Intelligent Information Infrastructure, D.-J. Deng, Y.-P. Lin, X. Yang, J. Zhu, Y.-B. Li, J. Luo, K.-C. Chen, IEEE Commun. Mag., 2017	Reviews 802.11ax MAC/PHY enhancements, including OFDMA-based random access, trigger frames for coordinated uplink scheduling, and spatial reuse, framed specifically around supporting dense, heterogeneous IoT-style device populations sharing WLAN infrastructure.	Argues that trigger-frame-based coordinated scheduling and spatial reuse are essential to sustaining efficiency as the number and heterogeneity of associated devices grows.	**Advantages:** Frames 802.11ax MAC design explicitly around heterogeneous, dense-device coexistence, directly relevant to WiFi-centric coordination goals. **Limitations:** A magazine-style overview without new quantitative evaluation of its own (*inferred limitation*).
12	Explicit Channel Coordination via Cross-Technology Communication, Z. Yin, Z. Li, S. M. Kim, T. He, MobiSys, 2018	Proposes using CTC not just for opportunistic interference avoidance but for explicit scheduling and reservation signaling between heterogeneous devices, allowing WiFi and ZigBee nodes, for example, to directly exchange channel-access intent and negotiate transmission windows.	Explicit CTC-based coordination reduces cross-technology collisions and improves channel utilization and fairness compared with implicit, sense-based coexistence schemes.	**Advantages:** Moves the field from passive interference avoidance to active, negotiated channel access; conceptually the closest prior work to CoWiC-MAC's coordinated MAC design goal. **Limitations:** Coordination overhead from CTC signaling exchange consumes airtime that could otherwise carry data, and reliability depends on the underlying CTC link quality (*inferred limitation*).
13	A Tutorial on IEEE 802.11ax High Efficiency WLANs, E. Khorov, A. Kiryanov, A. Lyakhov, G. Bianchi, IEEE Commun. Surv. Tutor., 2019	A comprehensive tutorial with analytical and simulation-based evaluation of the full 802.11ax MAC/PHY toolset, including OFDMA, MU-MIMO, spatial reuse/BSS coloring, and Target Wake Time, covering their combined performance in dense and heterogeneous WLAN deployments.	Shows that combining OFDMA scheduling with spatial reuse and TWT yields the largest efficiency gains in dense, heterogeneous deployments, though gains are sensitive to legacy-device presence and traffic mix.	**Advantages:** Offers one of the most complete, rigorously validated references on 802.11ax's coordinated-access mechanisms, useful as a MAC-design baseline. **Limitations:** Evaluation is confined to homogeneous 802.11ax/legacy-802.11 scenarios, not cross-radio-technology heterogeneity (*inferred limitation*).
14	Target Wake Time: Scheduled Access in IEEE 802.11ax WLANs, M. Nurchis & B. Bellalta, IEEE Wireless Commun., 2019	Explains and evaluates Target Wake Time (TWT), an 802.11ax mechanism in which the AP negotiates individual wake/access windows with stations, converting part of channel access from contention-based to scheduled, coordinated access.	TWT is shown to reduce contention overhead and improve energy efficiency, with particular benefit for IoT-class devices sharing a WLAN with regular high-throughput traffic.	**Advantages:** Demonstrates a standardized, AP-coordinated scheduling primitive directly analogous to the coordination concept the base paper generalizes across heterogeneous radios. **Limitations:** TWT coordination operates only within the 802.11 ecosystem and does not extend to non-WiFi heterogeneous devices (*inferred limitation*).
15	CDA: Coordinating Data Dissemination and Aggregation in Heterogeneous IoT Networks Using CTC, Y. Pan, S. Li, Y. Zhang, T. Zhu, J. Netw. Comput. Appl., 2020	Proposes a CTC-based protocol that coordinates simultaneous data dissemination (broadcast) and aggregation (collection) across WiFi and low-power heterogeneous IoT radios, addressing the inefficiency of running separate, uncoordinated dissemination and aggregation passes per technology.	Coordinated CTC-based dissemination and aggregation is shown to reduce completion time and overhead relative to running technology-specific dissemination and aggregation independently.	**Advantages:** Directly authored by the base paper's co-authors, Yan Pan and Shining Li, establishing a direct research lineage into CoWiC-MAC's WiFi-centric coordination approach. **Limitations:** Coordination logic is tailored to dissemination/aggregation traffic patterns rather than general MAC-level channel access (*inferred limitation*).
16	PolarScout: Wi-Fi Interference-Resilient ZigBee	Introduces a shell-shaping physical-layer modulation technique that reshapes the ZigBee signal constellation so that WiFi-	Shell-shaped ZigBee transmissions achieve substantially improved packet reception performance under	**Advantages:** A purely PHY-layer, receiver/transmitter-local fix requiring no cross-technology signaling or WiFi

S.No	Title of the Paper, Author Name, Year	Methodology	Key Findings	Advantage and Limitations
	Communication via Shell-Shaping, C. Shao, H. Park, H. Roh, W. Lee, H. Kim, IEEE/ACM Trans. Netw., 2020	induced interference is more easily distinguished and corrected at the receiver, improving ZigBee resilience without WiFi-side cooperation.	WiFi interference relative to standard O-QPSK ZigBee modulation.	modification. **Limitations:** Improves resilience but does not coordinate channel access, so collision probability under heavy WiFi load is only partially mitigated (*inferred limitation*).
17	Enabling WLAN and WPAN Coexistence via Cross-Technology Communication, S. Kim, Sensors, 2022	Proposes CTC-CSMA/CA, a bidirectional CTC mechanism that mimics WiFi's carrier-sense-multiple-access-with-collision-avoidance behavior between WiFi and ZigBee, allowing explicit channel-resource requests and feedback between the two technologies, plus beacon-synchronized low-duty-cycle ZigBee operation, implemented on commodity WiFi APs and ZigBee hardware with no 802.11 standard modification.	Experiments show CTC-CSMA/CA coordinates shared-channel access more accurately than conventional coexistence schemes, significantly improving ZigBee throughput as reported qualitatively by the authors.	**Advantages:** Fully standard-compliant and hardware-unmodified; explicitly reproduces CSMA/CA-style coordination across technologies, closely mirroring the coordinated, WiFi-centric MAC philosophy of CoWiC-MAC. **Limitations:** Validated on a two-technology (WiFi-ZigBee) testbed; scalability to additional heterogeneous radio types is not demonstrated (*inferred limitation*).
18	Enabling Global Cooperation for Heterogeneous Networks via Reliable Concurrent Cross Technology Communications, W. Chen, Z. Yin, T. He, IEEE Trans. Mob. Comput., 2022	Extends CTC from pairwise, point-to-point coordination to reliable, concurrent, multi-node cross-technology communication, addressing the scalability limitation of earlier CTC schemes when many heterogeneous devices must cooperate network-wide.	Demonstrates that reliable concurrent CTC sustains cooperation performance as the number of participating heterogeneous devices scales up, where pairwise CTC approaches degrade.	**Advantages:** Directly targets the network-wide, scalability gap left by earlier point-to-point CTC coordination schemes. **Limitations:** Increased signaling and coordination complexity is required for concurrency management (*inferred limitation*).
19	WibZig: Reliable and Commodity-Device Compatible PHY-CTC via Chip Emulation in Phase, T. Cheng, S. Li, F. Jiao, Y. Pan, IPSN, 2023	Proposes a phase-domain chip-emulation PHY-CTC technique, extending the WEBee-style emulation idea, that enables reliable WiFi-to-ZigBee communication on unmodified commodity hardware, improving robustness over amplitude/frequency-domain emulation approaches.	Reported to achieve improved reliability and commodity-hardware compatibility for PHY-layer CTC compared to prior emulation techniques, per the authors.	**Advantages:** Authored by the same research group as the base paper (Cheng, Li, Pan); WibZig is a direct technical precursor to CoWiC-MAC's WiFi-centric coordination mechanism. **Limitations:** Focused on point-to-point PHY-CTC reliability rather than full MAC-level coordinated channel access across a heterogeneous network (*inferred limitation*).
20	An Effective Data Dissemination Method in WAIC Networks Using Enhanced CTC, T. Cheng, S. Li, W. Dang, Y. Pan, Computer Networks, 2025	Proposes an enhanced CTC-based data dissemination method for Wireless Avionics Intra-Communications (WAIC) networks, a heterogeneous aviation wireless network context, building on the group's earlier PHY-CTC work to coordinate dissemination across heterogeneous nodes.	Reports improved dissemination effectiveness and efficiency over baseline CTC approaches in the heterogeneous WAIC network setting, per the authors.	**Advantages:** The most recent, directly preceding work by the base paper's own authors, forming the immediate research background for CoWiC-MAC. **Limitations:** Scoped to data-dissemination performance in the WAIC context rather than a general-purpose coordinated MAC mechanism, motivating the base paper's broader WiFi-centric MAC design (*inferred limitation*).

4. Literature Survey in Paragraph Format

Bianchi (2000) developed the analytical foundation for understanding how the IEEE 802.11 Distributed Coordination Function (DCF) behaves under contention. Rather than relying purely on simulation, the author modeled the binary exponential backoff process as a two-dimensional discrete-time Markov chain and derived closed-form expressions for saturation throughput under both the Basic Access and RTS/CTS mechanisms, assuming an ideal channel and a finite number of contending stations. The derived throughput values matched simulation results very closely, and the analysis showed that RTS/CTS becomes increasingly

advantageous over Basic Access as the number of contending stations and frame sizes grow, since it lowers the cost of a collision. This work is significant because it gave the research community its first rigorous tool for reasoning about WiFi's contention-based channel access, and later MAC studies, including those on heterogeneous coexistence, still build on this Markov chain framework. Its main limitation is that it only models a single, homogeneous WiFi network under saturated, ideal-channel conditions, so it says nothing about coexistence with other radio technologies.

Jamieson and Balakrishnan (2007) tackled a different inefficiency in wireless MAC design: a single bit error forces an entire 802.11 frame to be discarded and retransmitted. Their system, PPR (Partial Packet Recovery), lets the receiver use physical-layer confidence information to identify exactly which chip or bit segments of a corrupted frame arrived correctly, then requests retransmission of only the corrupted portions. Because this avoids resending data already received successfully, PPR achieved substantially higher throughput than standard ARQ retransmission as channel loss rates increased. The scheme's chief advantage is that it recovers airtime that would otherwise be wasted on full-frame retransmission, without requiring any change to higher-layer protocols. Its limitation is that it depends on receiver hardware capable of estimating bit-level confidence, which adds implementation complexity and offers diminishing benefit once error rates are already low.

Huang, Xing, Zhou, and Zhou (2010) addressed ZigBee sensor traffic being crowded out by WiFi transmissions sharing the same 2.4 GHz band. Instead of modifying WiFi devices, the authors built a system that senses and predicts the idle temporal white space that naturally occurs between WiFi transmissions, then schedules ZigBee transmissions into those predicted gaps. Because this approach requires no cooperation from the WiFi side, it can be deployed unilaterally on the low-power network alone. Experiments showed that scheduling around predicted WiFi-idle periods substantially improved ZigBee packet delivery compared to letting the two networks interfere without management. The main strength of this design is its WiFi-transparent deployability; the drawback is that prediction accuracy falls off when WiFi traffic becomes bursty or unpredictable, since coordination is one-directional rather than negotiated between the two technologies.

Liang, Priyantha, Liu, and Terzis (2010) took a coding-based approach to the same coexistence problem, focusing on how ZigBee networks can survive WiFi interference rather than avoid it. They designed a forward-error-correction scheme with redundancy structured specifically around the burst-loss pattern that WiFi interference tends to produce in ZigBee frames, so a receiver can still recover a packet even when part of it has been corrupted. Because the coding lives entirely on the ZigBee side, no WiFi-side cooperation is needed. Their evaluation showed markedly higher packet reception rates than uncoded transmission under realistic WiFi interference conditions. The advantage of this scheme is that it can be deployed unilaterally on resource-constrained ZigBee nodes; the tradeoff is that the added redundancy reduces effective payload throughput and increases energy consumption, and the technique tolerates interference rather than eliminating it.

Zhang and Shin (2011) proposed one of the earliest general mechanisms for enabling heterogeneous WiFi and ZigBee devices to coexist without a shared MAC protocol. Their approach uses physical-layer energy and activity detection so each technology can sense when the other is transmitting and time its own transmissions to avoid overlap, rather than requiring protocol-level interoperability. Experiments demonstrated substantially lower collision-induced packet loss for both ZigBee and WiFi compared to uncoordinated coexistence. This work is notable for establishing that cross-technology coexistence does not

require a common MAC standard, only mutual activity awareness. However, because coordination here is implicit and detection-based rather than explicitly signaled, its precision and scalability beyond two coexisting technologies remain limited.

Wang, Wang, Zheng, Zeng, Zheng, and Zhang (2014) extended the white-space idea to safety-critical medical body-area networks with WiCop. Their system passively observes WiFi traffic to build a model of its temporal occupancy pattern, then schedules wireless body-area network (WBAN) transmissions into the predicted gaps while providing worst-case latency guarantees, a requirement that ordinary best-effort coexistence schemes cannot meet. The evaluation showed WiCop could bound worst-case access latency for medical sensor traffic even while sharing spectrum with an unmodified WiFi network. Its principal advantage is offering a deterministic safety guarantee suited to medical applications rather than a probabilistic one. The main limitation is that scheduling accuracy depends on correctly predicting short-term WiFi traffic behavior, which becomes harder in highly dynamic or multi-access-point environments.

Guo, Cao, Zhang, and Liu (2014) revisited the coding approach to WiFi-ZigBee coexistence but made the redundancy adaptive rather than fixed. Their scheme continuously estimates the intensity of WiFi interference in real time and adjusts the forward-error-correction code rate accordingly, so ZigBee nodes use only as much redundancy as current channel conditions require, avoiding the throughput penalty of always assuming worst-case interference. Their results showed the adaptive scheme outperformed both static-redundancy coding and uncoded transmission across a range of WiFi traffic loads. The benefit of this design is interference-proportional efficiency; the cost is that continuously estimating interference levels adds computational and energy overhead on already resource-constrained ZigBee nodes.

Yang, Yan, Li, Zhang, Tao, and You (2016) built an empirical, statistical model of cross-technology interference between WiFi and ZigBee and paired it with an active mitigation strategy combining interference-aware channel and timing selection with adaptive transmission. Unlike schemes that only tolerate interference after it occurs, this approach tries to characterize and then actively avoid it. The authors validated their interference model experimentally and showed the resulting mitigation strategy achieved a substantially improved ZigBee packet reception ratio with fewer retransmissions under realistic interference conditions. The main strength of the work lies in combining interference modeling with active mitigation for greater robustness. A limitation is that the model's calibration is specific to the WiFi-ZigBee interference regime studied and may need retuning for other technology pairs or channel conditions.

Bellalta (2016) authored a survey-style overview of the mechanisms introduced by the then-emerging IEEE 802.11ax amendment, including OFDMA-based multi-user access, MU-MIMO, and spatial reuse through BSS coloring, all aimed at improving efficiency in dense, overlapping basic-service-set deployments. As a tutorial rather than an experimental study, the paper reports no new measurements of its own; instead, it consolidates and explains the new MAC and PHY toolbox for the research community. The discussion identifies OFDMA scheduling and spatial reuse as the mechanisms expected to deliver the largest efficiency gains, while flagging legacy-device coexistence as an unresolved challenge. Its value lies in providing an accessible, comprehensive reference to coordinated-access mechanisms analyzed in more depth by later authors; its limitation is simply that it offers no original experimental validation.

Li and He (2017) introduced WEBee, a technique that lets a WiFi radio communicate directly with a ZigBee radio at the physical layer through emulation rather than through packet-level signaling. By manipulating only the payload of standard WiFi OFDM packets, a high-speed WiFi transmitter can emulate the exact

waveform a ZigBee receiver expects, all without any hardware or firmware modification on either side. This solved the throughput bottleneck of earlier cross-technology communication (CTC) schemes, which operated at only tens of bits per second. WEBee achieved over 99 percent reliable parallel communication between WiFi and ZigBee at 126 Kbps in noisy environments, roughly 16,000 times faster than prior packet-level modulation approaches. Its advantage is a dramatic, hardware-free throughput gain that made PHY-layer CTC practical, and it went on to underpin later coordination protocols, including the base paper's own earlier WibZig work. Its limitation is that emulation fidelity depends on precise timing and waveform control at the WiFi transmitter, and the technique was validated mainly in the WiFi-to-ZigBee direction.

Deng, Lin, Yang, Zhu, Li, Luo, and Chen (2017) reviewed IEEE 802.11ax's MAC and PHY enhancements with a specific emphasis on supporting dense, heterogeneous IoT-style device populations rather than conventional WiFi traffic alone. The paper walks through OFDMA-based random access, trigger-frame-based coordinated uplink scheduling, and spatial reuse, arguing these mechanisms are essential for sustaining efficiency as both the number and heterogeneity of associated devices increase. Being a magazine-style review, it presents no new quantitative results, but it frames the evolution of 802.11ax MAC design explicitly around heterogeneous-device coexistence, which makes it directly relevant to WiFi-centric coordination goals such as those pursued by the base paper. Its limitation, as with similar surveys, is the absence of original experimental evaluation.

Yin, Li, Kim, and He (2018) pushed cross-technology communication a step further than prior emulation work by using it for explicit channel coordination rather than opportunistic interference avoidance. In their design, heterogeneous devices such as WiFi and ZigBee nodes directly exchange channel-access intent through CTC signaling and negotiate transmission windows with each other, rather than simply sensing and avoiding one another's traffic. This explicit negotiation reduced cross-technology collisions and improved channel utilization and fairness compared to implicit, sensing-based coexistence schemes. Conceptually, this is the closest piece of prior work to CoWiC-MAC's own coordinated MAC design goal, since it treats coordination as an active, negotiated process rather than a passive one. Its limitation is that the CTC signaling exchange itself consumes airtime that could otherwise carry data, and the scheme's reliability depends on the quality of the underlying CTC link.

Khorov, Kiryanov, Lyakhov, and Bianchi (2019) produced one of the most complete tutorials on IEEE 802.11ax, combining analytical modeling with simulation to evaluate OFDMA, MU-MIMO, spatial reuse through BSS coloring, and Target Wake Time together rather than assessing each mechanism separately. This joint evaluation approach revealed that combining OFDMA scheduling with spatial reuse and Target Wake Time produces the largest efficiency gains in dense and heterogeneous deployments, though the size of those gains is sensitive to the presence of legacy devices and to the traffic mix. As a rigorously validated reference on 802.11ax's coordinated-access mechanisms, this tutorial is a useful baseline for MAC design work. Its scope, however, is confined to homogeneous 802.11ax and legacy 802.11 scenarios rather than coexistence across different radio technologies.

Nurchis and Bellalta (2019) focused specifically on Target Wake Time (TWT), the 802.11ax mechanism that lets an access point negotiate individual wake and access windows with each station, effectively converting part of channel access from contention-based to AP-coordinated scheduling. Their evaluation showed TWT reduces contention overhead and improves energy efficiency, with the greatest benefit for IoT-class devices that must share a WLAN with regular, high-throughput traffic. TWT is significant because it demonstrates a

standardized, AP-coordinated scheduling primitive directly analogous to the kind of coordination the base paper later generalizes across heterogeneous radio types. Its limitation is that TWT coordination operates entirely within the 802.11 ecosystem and cannot, by itself, extend to non-WiFi heterogeneous devices.

Pan, Li, Zhang, and Zhu (2020) proposed CDA, a protocol that uses cross-technology communication to coordinate data dissemination and aggregation together across WiFi and low-power heterogeneous IoT radios, rather than running these two operations separately for each technology. This coordinated approach reduced overall completion time and overhead compared to running technology-specific dissemination and aggregation passes one after another. CDA is particularly relevant to CoWiC-MAC because two of its authors, Yan Pan and Shining Li, also co-authored the base paper, making this work a direct predecessor in the same research lineage. Its limitation is that the coordination logic is tailored specifically to dissemination and aggregation traffic patterns rather than to general-purpose MAC-level channel access.

Shao, Park, Roh, Lee, and Kim (2020) approached WiFi-ZigBee coexistence from the physical layer with PolarScout, a technique that reshapes the ZigBee signal constellation, described by the authors as shell-shaping, so WiFi-induced interference becomes easier for the receiver to distinguish and correct. Because the modification lives entirely at the ZigBee transmitter and receiver, no cooperation from the WiFi side is required. Their evaluation showed substantially improved packet reception under WiFi interference compared to standard O-QPSK ZigBee modulation. The strength of this approach is that it is a purely local, PHY-layer fix needing no cross-technology signaling. Its limitation is that it improves resilience to interference without coordinating channel access, so under heavy WiFi load the probability of collision is only partially reduced rather than eliminated.

Kim (2022) proposed CTC-CSMA/CA, a mechanism that uses bidirectional cross-technology communication to mimic WiFi's own carrier-sense-multiple-access-with-collision-avoidance behavior between WiFi and ZigBee. In this design, ZigBee devices can explicitly request channel resources and receive feedback from WiFi, and additionally synchronize to WiFi's periodic beacon to operate at a low duty cycle. Implemented entirely on commodity WiFi access points and ZigBee hardware, the scheme requires no modification to the 802.11 standard. Experiments showed CTC-CSMA/CA coordinates shared-channel access more accurately than conventional coexistence schemes and significantly improves ZigBee throughput. Because it explicitly reproduces CSMA/CA-style coordination across two different technologies using only commodity hardware, this work closely mirrors the coordinated, WiFi-centric MAC philosophy that CoWiC-MAC pursues. Its limitation is that the evaluation was confined to a two-technology testbed, and scalability to additional heterogeneous radio types was not demonstrated.

Chen, Yin, and He (2022) addressed a scalability gap left by earlier cross-technology communication work: most prior CTC schemes coordinate only pairs of devices at a time. Their scheme extends CTC to reliable, concurrent, multi-node communication, allowing many heterogeneous devices to cooperate across the network simultaneously rather than one link at a time. Their results demonstrated that this concurrent design sustains cooperation performance as the number of participating heterogeneous devices scales up, in contrast to pairwise CTC approaches, which degrade under the same conditions. The main advantage of this work is that it directly targets the network-wide scalability limitation of point-to-point CTC coordination. A limitation is that supporting concurrency introduces added signaling and coordination complexity that pairwise schemes do not need to manage.

Cheng, Li, Jiao, and Pan (2023) presented WibZig, a physical-layer cross-technology communication technique that extends the WEBee-style emulation approach into the phase domain through chip emulation, improving reliability over earlier amplitude and frequency-domain emulation methods while remaining compatible with unmodified commodity hardware. The authors reported improved reliability and hardware compatibility for PHY-layer CTC relative to prior emulation techniques. This paper carries particular weight for the present survey because it was authored by the same research group behind CoWiC-MAC, namely Cheng, Li, and Pan, making WibZig a direct technical precursor to the base paper's own coordination mechanism. Its limitation is that it focuses on point-to-point PHY-layer CTC reliability rather than on full MAC-level coordinated channel access across an entire heterogeneous network.

Cheng, Li, Dang, and Pan (2025) applied the group's earlier PHY-CTC techniques to a new setting, Wireless Avionics Intra-Communications (WAIC) networks, proposing an enhanced cross-technology-communication-based method for disseminating data across heterogeneous nodes in this aviation network context. The authors reported improved dissemination effectiveness and efficiency relative to baseline CTC approaches within the WAIC setting. As the most recent publication by the base paper's own authors prior to CoWiC-MAC itself, this paper forms the immediate research background from which the base paper emerges. Its limitation is that it remains scoped to data-dissemination performance within the WAIC context rather than proposing a general-purpose coordinated MAC mechanism, a gap the base paper goes on to address directly.

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5. Relevance of the Literature Survey to CoWiC-MAC

Common problems addressed. The surveyed papers converge on a single structural problem. WiFi and low-power, technologically incompatible radios, such as ZigBee/802.15.4 and other WPAN/IoT devices, must share the unlicensed 2.4 GHz ISM band, but neither a common MAC protocol nor mutual protocol awareness exists between them. This produces cross-technology interference, unpredictable packet loss, and inefficient, uncoordinated channel access: the same heterogeneous-coexistence problem that motivates CoWiC-MAC.

Common techniques used. Three technique families recur across the survey. The first is passive or implicit coexistence, which senses WiFi's temporal white space or activity to schedule the other technology's transmissions around it (Huang 2010; Liang 2010; Zhang & Shin 2011; Wang 2014; Shao 2020). The second is Cross-Technology Communication (CTC), which enables direct signaling between heterogeneous radios and has evolved from physical-layer emulation (Li & He 2017; Cheng et al. 2023) to explicit coordination and reservation protocols (Yin et al. 2018; Kim 2022; Chen et al. 2022; Pan et al. 2020; Cheng et al. 2025). The third is WiFi-native coordinated scheduling: mechanisms built into the 802.11ax amendment itself, such as OFDMA, spatial reuse, and Target Wake Time, that convert part of WiFi's own channel access from contention-based to AP-coordinated (Bellalta 2016; Deng 2017; Khorov 2019; Nurchis & Bellalta 2019). All three families ultimately rest on Bianchi's (2000) original DCF contention model and on Jamieson and Balakrishnan's (2007) link-layer resilience techniques.

Remaining limitations in existing approaches. Passive and implicit schemes avoid interference reactively but cannot negotiate access precisely and scale poorly beyond two technologies. Early CTC work in the style of WEBee achieves high-throughput cross-technology signaling but was largely demonstrated point-to-point,

and pairwise CTC coordination (Yin 2018; Kim 2022) does not scale efficiently to many concurrent heterogeneous devices, a gap only partly closed by concurrent-CTC work (Chen et al. 2022). Meanwhile, 802.11ax's coordinated-access mechanisms, such as TWT and OFDMA, are powerful but operate strictly within the WiFi ecosystem and do not extend coordination to non-WiFi heterogeneous radios. No single prior scheme unifies WiFi-native coordinated scheduling with cross-technology signaling into one MAC mechanism explicitly centered on WiFi as the coordinating hub.

Research background for CoWiC-MAC. The base paper's own reference list shows a direct lineage: Pan et al. (2020, CDA) and Cheng et al. (2023, WibZig; 2025, WAIC-CTC) are prior work by the same author group, progressively building CTC-based coordination, first for data dissemination and aggregation, then for reliable PHY-layer emulation, then for heterogeneous avionics dissemination. CoWiC-MAC generalizes this trajectory into a full coordinated, WiFi-centric MAC mechanism, positioning WiFi, the higher-capability, infrastructure-anchored radio, as the active coordinator of channel access for heterogeneous low-power devices, rather than treating coexistence as passive avoidance or point-to-point signaling.

What makes CoWiC-MAC relevant compared with prior work. Unlike passive white-space or interference-tolerant coding schemes, CoWiC-MAC targets active coordination rather than tolerance. Unlike pairwise CTC schemes, it is explicitly WiFi-centric and MAC-level, aiming at network-wide coordination rather than single-link signaling. And unlike 802.11ax's TWT/OFDMA mechanisms, which coordinate only WiFi stations, CoWiC-MAC's WiFi-centric design extends coordinated medium access across heterogeneous device types by combining the scheduling discipline of 802.11ax-style coordination with the cross-technology reach of CTC, addressing the scalability and cross-technology-scope gaps identified across the survey.

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7. Base Paper Reference

Base Paper

Cheng, T., Li, S., & Pan, Y. "CoWiC-MAC: A Coordinated WiFi-Centric MAC Mechanism in Heterogeneous Wireless Networks." *Computer Networks*, 287, Art. 112479, 2026. DOI: 10.1016/j.comnet.2026.112479.